

Spot an Error

And WIN Award!

You can win one thousand rupees plus a certificate from the editor of Electronics For You magazine if you spot and report any factual error in the latest issue of the magazine. Your report will be published in a coming issue of the magazine, giving credit to you.

Though every care is taken in preparation of the content for the magazine, an error can slip in at times. So we invite our learned readers to point out any error that they spot, for the benefit of the other readers.

Rules:

- The error has to be reported before end of the month that is mentioned on cover of the magazine
- The report has to be sent in writing or by email addressed to the Editor, EFY Magazine (editsec@efyindia.com)
- The error should be of factual nature, in the information published in the magazine, and not a grammatical or syntax error
- The correct information should also be provided along with the error spotted by you
- In case more than one person reports the same error, the one whose report reaches us the earliest will get the award money
- In case two or more persons jointly report an error, the award money will be divided amongst them
- EFY employees, their family members and friends, and EFY associates and their family members and friends are not eligible
- Decision of the editor will be final in this matter

electronics
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ing rise to challenges such as isolation of noise from neighbours. Measuring simply crosstalk and distortions is a thing of the past; today, people want to know the psycho-acoustical perception characteristics of the sound, and this has to be tested!

Similarly, on the

video front, HD video is transferred to multiple locations, bringing in the need for a high-bandwidth wired or wireless medium. HDMI and its variants are extensively used in automobiles.

Latest test suites help in generating and testing a variety of digital audio and video standards by using a graphical system design approach. Each analyser/generator combines high-performance, PXI-based modular instruments with configuration-based software to simplify testing of multimedia devices such as set-top boxes, Blu-ray players and LCD displays. Using a single PXI system, engineers can test CVBS, S-Video, CAV, VGA and HDMI 1.4 interfaces along with audio interfaces to optimise system performance.

Other aspects to be tested

There are also sensors that enable keyless entry, voice-activated commands, and driver and passenger safety. PXI cards that can work with high-level programming software help perform multiple tests on these components.

Integrating test systems with product lifecycle management

In recent years, the growing complexity of automotive products, shorter product development lifecycles and increasingly stringent compliance standards have led to the need for formalised requirements in product development and testing.

An effective, cost-efficient qual-

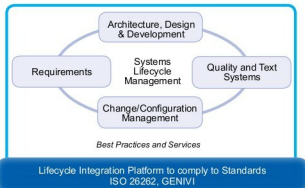


Fig. 2: System lifecycle management (Courtesy: National Instruments India)

ity and testing strategy must be based on clearly-defined requirements. Requirements help you specify the exact functionality of a product and guarantee that it is tested correctly and completely. Furthermore, compliance standards emphasise the need to prove links between higher-level and lower-level requirements and the implementation of requirements in the product. Proper documentation and reporting is also essential at all stages. So engineers are increasingly using requirements management tools to make the process of storing and analysing the requirements cost-efficient, and to ensure that the testing of automotive electronics is completely integrated with the rest of the product lifecycle.

Such tools completely link the development and verification documents to formal requirements stored in documents and databases, perform coverage and impact analysis, graphically display coverage relationships, and generate comprehensive reports.

To sum up

Test engineers in the automotive industry are challenged to increase test system performance while decreasing the system size to reduce the total cost of design, production and validation test. Conscious switch to a modular, software-defined approach is helping engineers to realise these goals. ●

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